KV6014 CHALLENGE 5 · MASTER UX USER FLOW & DECISION STATE MACHINE (v2.0 — 100% UI & DEV SYNCED)

ROSE Event Management App — 3-Role + US-09 End-to-End Flow (SCR-01–SCR-12 · 49 States)

Synced with frontend/live/*.html (R1), GitHub Project #1 Issues #6–#37 (R2), and US-09 / US-10 (R3)

BACKEND DB ENUM: users.role IN ('user', 'leader', 'organizer')

- 3 Core System Roles: Role 1 Participant ('user') · Role 2 Leader ('leader') · Role 3 Organizer ('organizer')
- Role 3B (US-09 Event-Day Check-In & Walk-In) is a Sub-Function under Role 3 ('organizer' · Zero 4th Role)

FLOWCHART & PROTOTYPE NOTATION LEGEND:

Start/End · SCR / State · Decision · API / DB · P/L/O/F = 49 States

P-01..P-14+P-06E (15) · L-01..L-12 (12) · O-01..O-12 (12) · F-01..F-10 (10)

ROLE 1 · PARTICIPANT (DB: role="user" / Guest · States P-01 to P-14 + P-06E)

Live R1 Pages: index.html, event.html, signup.html, login.html, registration.html

Supports zero-account Guest SMS OTP booking (#34) OR Optional Member Sign-Up/Login (#31 role='user' guard).

START: Discovery
Web / ?ref= / Flyer

SCR-01 · index.html [P-01..02]
Search & Filter Clinics
Keyword #event-keyword · US-06
GET /api/v1/events

SCR-02 · event.html [P-03]
Clinic & Dual Quotas
Online (18) vs Offline (12) slots
GET /api/v1/events/{id}

online_quota > 0 Left?

Yes

SCR-06 · registration [P-05,06E]
7-Field ParticipantCreate
Name, Gender, DOB, Phone, Addr, MyKad
POST /api/v1/participants (>=16px)

SCR-07A · Issue #34 [P-06..07]
6-Digit SMS OTP Verify
Guest phone ownership check
POST /api/v1/registrations

SCR-07B · CONFIRMED [P-08]
Receipt #ROSE-882 + QR
Atomic lock: online_quota - 1
Server-Authoritative Pass

EVENT DAY · US-09 Check-In
Attend & HPV Self-Swab
Present QR / MyKad at Station
Checked in via SCR-11 (F-04)

END: Lab SMS [P-14]
Result & Follow-up Care
status='CHECKED_IN'

SCR-03 signup.html & SCR-04 login.html [P-04 · US-01 · #31 Guard]
Optional Account Sign-Up / Login (or Continue as Guest)
POST /api/v1/auth/register (locks role='user') · POST /api/v1/auth/login

Full (0 Left)

SCR-07D · Issue #37 [P-11..12]
Join Waitlist Queue #1..N
Auto-queued when online_quota=0
status='WAITLISTED' (US-08)

Slot Released?
(Issue #36 -> #37)

P-13: Claim <24h

If Cancels

SCR-07C · Issue #36 [P-09..P-10]
2-Step Cancellation Guard Modal
Prevents accidental drops · Releases slot
DELETE /api/v1/registrations/{id}

Auto-promotes Waitlist #1 (US-08 · Issue #37)

Clicks ?ref= link

Enters Confirmed Roster

ROLE 2 · COMMUNITY LEADER (DB: role="leader" · States L-01 to L-12)

Screens: SCR-04 (L-01), SCR-08 Outreach Hub (L-02..04), SCR-09 OCR & Flyers (L-05..12)

Manages reserved village offline_quota (20 slots), ?ref= attribution, Split-Screen Paper OCR, & Self-Reported Flyer logs.

START: Sign In [L-01]
SCR-04 · role='leader'

SCR-08 · Leader Hub [L-02]
Village Offline Quota
Reserved offline_quota: 20 slots
US-03 · Issue #31 & #32

Outreach / Intake Mode?

Digital (?ref=)

SCR-08 · Referral Share [L-03]
Copy WhatsApp Invite Link
?ref=LDR-AISYAH (Verified Track)

Flyers

SCR-08 · Flyer & AI Draft [L-04]
Print PDF / Request AI Flyer
US-04: PENDING_ORGANIZER_APPROVAL

SCR-09C · US-05.3 [L-10..L-12]
Log Physical Flyers (1–500)
Tag: Self-Reported Reach (0 Quota Deduct)

Proxy / Paper OCR

SCR-09A · US-07 [L-05..L-07]
Split-Screen Paper OCR Scan
Left: Bounding Box · Right: 7-Field Form
Amber Alert (<85% Conf NRIC/DOB)

Human Verify Lock Ticked?
[L-07 -> L-08]

Verified OK

SCR-09B · Confirmed [L-09]
Proxy Intake Created
Deducts village offline_quota - 1
POST /api/v1/participants

⚡ Live Demand Balancing Directive Banner on SCR-08 (US-05 / Issue #32)

- When session reaches >=80% (82% in demo), Organizer dispatches a 1-Click [Pause Outreach] directive.
- Community Leader dashboard updates immediately to halt new flyer distribution and avoid crowding.
- **If turnout <40%, Organizer sends [Boost Outreach] directive to release extra village slots.**

O-08: Trigger 1-Click [Pause Outreach] Signal (US-05.2)

ROLE 3 ORGANIZER AUTH SCOPE (role='organizer' · US-09 SUB-FUNCTION)

ROLE 3A · ROSE ORGANIZER — PRE/POST-EVENT CONSOLE (role="organizer" · States O-01 to O-12)

Screens: SCR-04B 403 Guard (O-01), SCR-05 create-event.html (O-03..06), SCR-10 (O-07..10), SCR-12 (O-11..12)

Schedules dual-quota clinics, monitors T-05-01 KPIs, sends >=80% Pause signals, & reconciles closeout.

START: Auth [O-01..02]
Bearer JWT role='organizer'
403 Guard: #organizer-prompt

SCR-05 · create-event [O-03..06]
Schedule & Dual Quotas
online_quota: 30 | offline_quota: 20
POST /api/v1/events (US-02)

US-03 / US-04 · Governance
Assign Village Leaders
Allocate offline slots & review
AI flyer drafts (Issue #32)

SCR-10 · Dashboard [O-07]
Live Demand Monitor
T-05-01 KPIs: 41/50 (82% Cap)
US-05 · Issue #32 & #33

Capacity >= 80% (82% Alert)?

SCR-10 · US-06.2 [O-09..10]
OCR & Eligibility Queue
Approve flagged proxy records
PDPA Masked: ******-01-5832

SCR-12 · Closeout [O-11 · US-10]
Event Closeout & Attribution
Reconcile 50 Cap vs 41 Checked-In (82%)
Verified Bookings vs Unverified Flyers

END: Export CSV [O-12]
Turnout & Leader ROI
PDPA Anonymised Report

Top Nav [Field Check-In] Hotspot (O-07 / O-10 -> F-01): Launch US-09 Event-Day Sub-Function

F-10 -> O-11: Feeds 41 Arrivals

ROLE 3B [US-09 SUB-FUNCTION] · EVENT-DAY CHECK-IN & WALK-IN (States F-01 to F-10)

Auth: role='organizer' (US-09 · Zero 4th DB Role)

Screens: SCR-11A Roster (F-01..04,09), SCR-11B Walk-In (F-05..06), SCR-11C 409 (F-08)

START: Preload [F-01..02]
Cache 41 Roster Records
GET /events/{id}/roster

SCR-11A · Check-In [F-03]
Search Cached Roster
Filter by Name, MyKad, or #ROSE
0ms Offline IndexedDB Lookup

Pre-Registered on Roster?

Yes

SCR-11A · 1-Tap Check-In [F-04]
Queue Arrival Locally
Verify MyKad last-4 digits
IndexedDB (op_chk_884)

No (Walk-In)

SCR-11B · Walk-In Modal [F-05..F-06]
Register Walk-In (US-09 / FR-05)
7-Field Intake · Consumes offline_quota - 1
Queued locally with arrival check-in

PENDING_SYNC QUEUE [F-04..07]
Persistent Offline Banner
Zero fake server confirmation
POST /api/v1/checkins/sync

HTTP 409 Sync Conflict?

No (Clean)

SCR-11A · SYNCED [F-09..F-10]
Arrival Confirmed in DB
PostgreSQL updated (41 Arrivals)
0 Pending · 0 Unresolved

Yes (Cancelled while offline)

SCR-11C · HTTP 409 Conflict Modal [F-08]
Explicit Operator Adjudication (#Q-103)
Compare Local Check-In (09:41) vs Cloud Cancel (08:55)
[Re-Instate & Check In] OR [Honor Cancellation]

Resolved -> F-09

KV6014 ARCHITECTURAL & UI PROTOTYPE ALIGNMENT SUMMARY (v2.0 — LOG TASKS 1, 2 & 3 SYNCHRONISED)

- Database Schema Truth: backend/src/models/users.py strictly defines 3 roles: users.role IN ('user', 'leader', 'organizer'). Event-day check-in & walk-in intake (US-09) runs as Role 3B under role='organizer'.
- Full 12-Screen & 49-State Coverage: SCR-01..SCR-06 (6 Live HTML/ESM Pages) + SCR-07..SCR-10 (R2 Issues #31–#37) + SCR-11..SCR-12 (R3 US-09 & US-10) · Forest Green (#1F5547) Design System.